



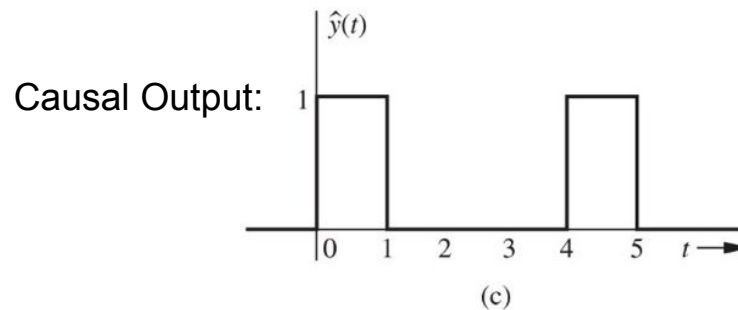
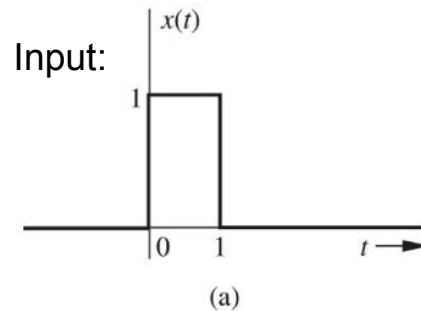
SYDE252 - W03B - System Response

Dr. Matt Borland

System Properties

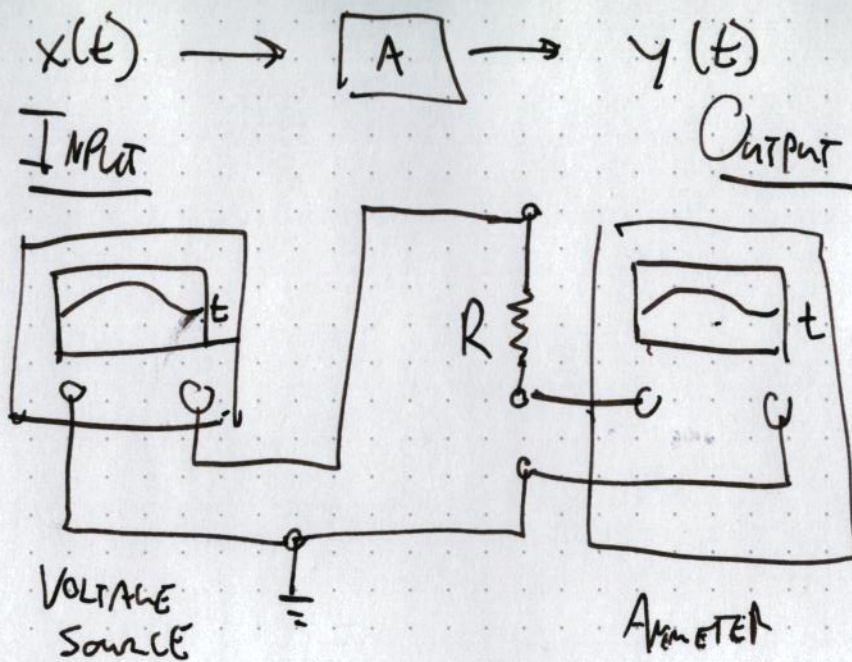
Causal Systems

The value of the output at the present instant depends only on the past and present values of the input, not its future values. These can exist in real-time!

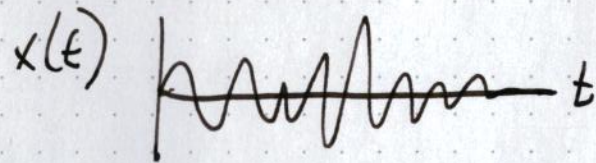


CAUSAL EXAMPLES?

- Ignoring Prof's QUESTION
- PUNCHING A BAG.
- RESISTOR
- CAR SUSPENSION



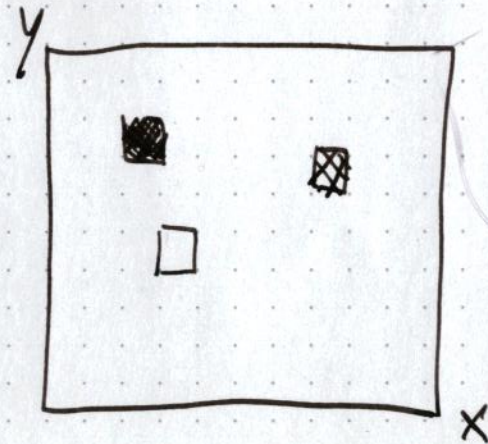
WHAT'S A SIGNAL?



VALUE THAT CHANGES WITH
TIME 1-DIMENSIONAL

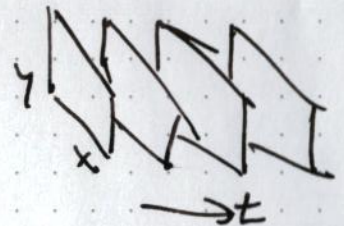
IF WE PRE-RECORD A SIGNAL WE
CAN CALCULATE THE CURRENT OUTPUT
VALUE AS A FUNCTION OF FUTURE VALUES...

2D SIGNAL



$P(x, y)$

3D? $V(x, y, t)$



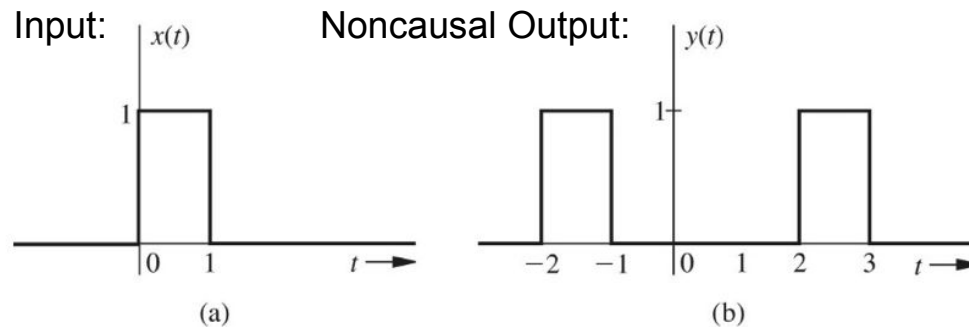
System Properties

Causal Systems

Examples?

System Properties

Systems that don't do this are **Noncausal!**



These can't exist in real-time!

...we could record a signal, then “post-process” it...

Sounds, images, video, etc...

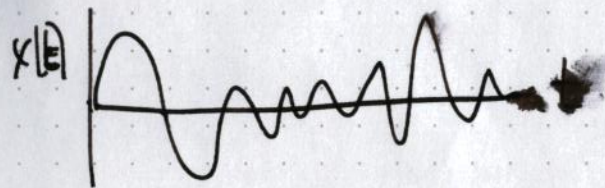
System Properties

Non-Causal Systems

An image is a signal?

W03B

What's a signal?



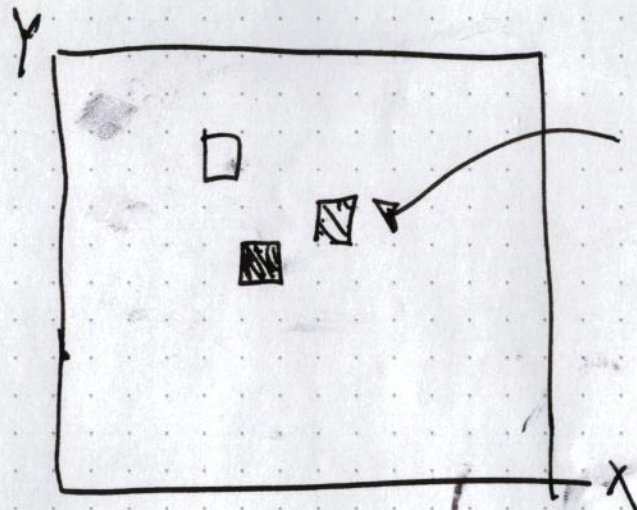
VALUE THAT CHANGES WITH TIME

1-DIMENSIONAL SIGNAL!

NON-CAUSAL SYSTEMS DON'T NEED TO WAIT FOR THE INPUT SIGNAL TO HAPPEN BEFORE YOU SEE AN OUTPUT RESPONSE

YOU COULD POST-PROCESS A TIME-DOMAIN SIGNAL AND DETERMINE THE CURRENT OUTPUT VALUE ON FUTURE INPUT VALUES...

$$y[t] = (x[t+1])^2$$

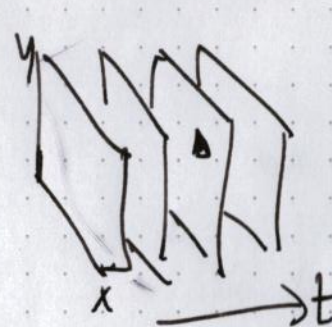


2 DIMENSIONAL SIGNAL.

AN IMAGE IS A SIGNAL!

$I(x,y)$

WHAT ABOUT VIDEO?



$V(x,y,t)$

System Properties

Instantaneous Systems

The value of the output at the present instant depends only on the present values of the input. The past history of the system is irrelevant. These systems are also called **memoryless**. If the system does have memory it is called **Dynamic** – maybe some energy is stored???

Instantaneous: $y(t - 1) = 2x(t - 1)$

Dynamic:

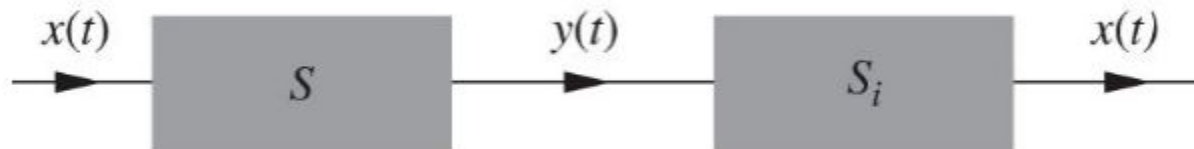
$$y(t - 1) = 2x(t - 1) - x(t - 2)$$

System Properties

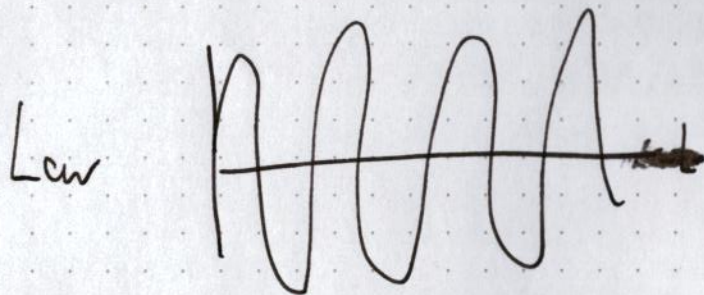
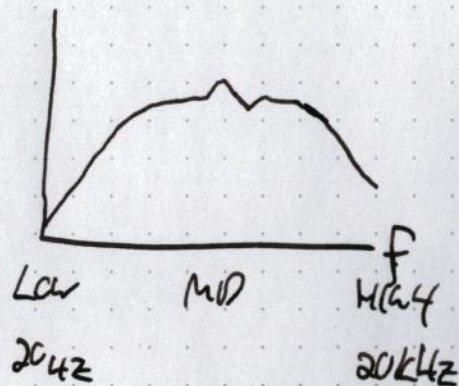
Invertible Systems

For a given system S , if we can obtain the input $x(t)$ back from the corresponding output $y(t)$ by some operation, the system S is said to be invertible.

If we can't get our input back by some operation, then it is **noninvertible**.

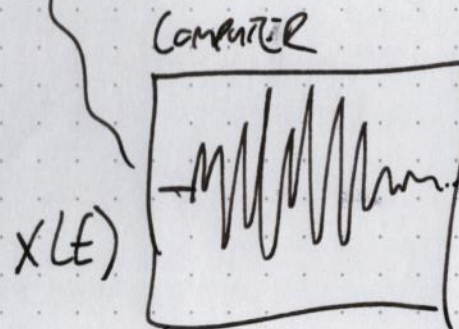
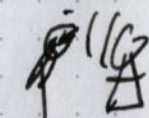
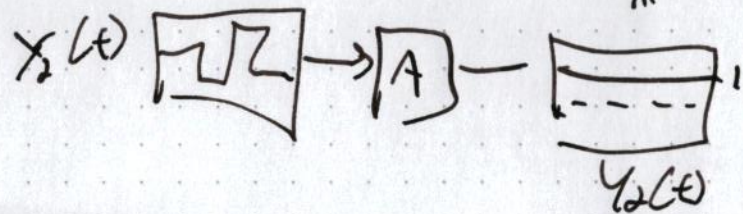
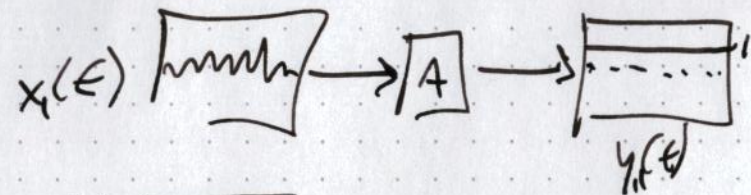


https://en.wikipedia.org/wiki/RIAA_equalization



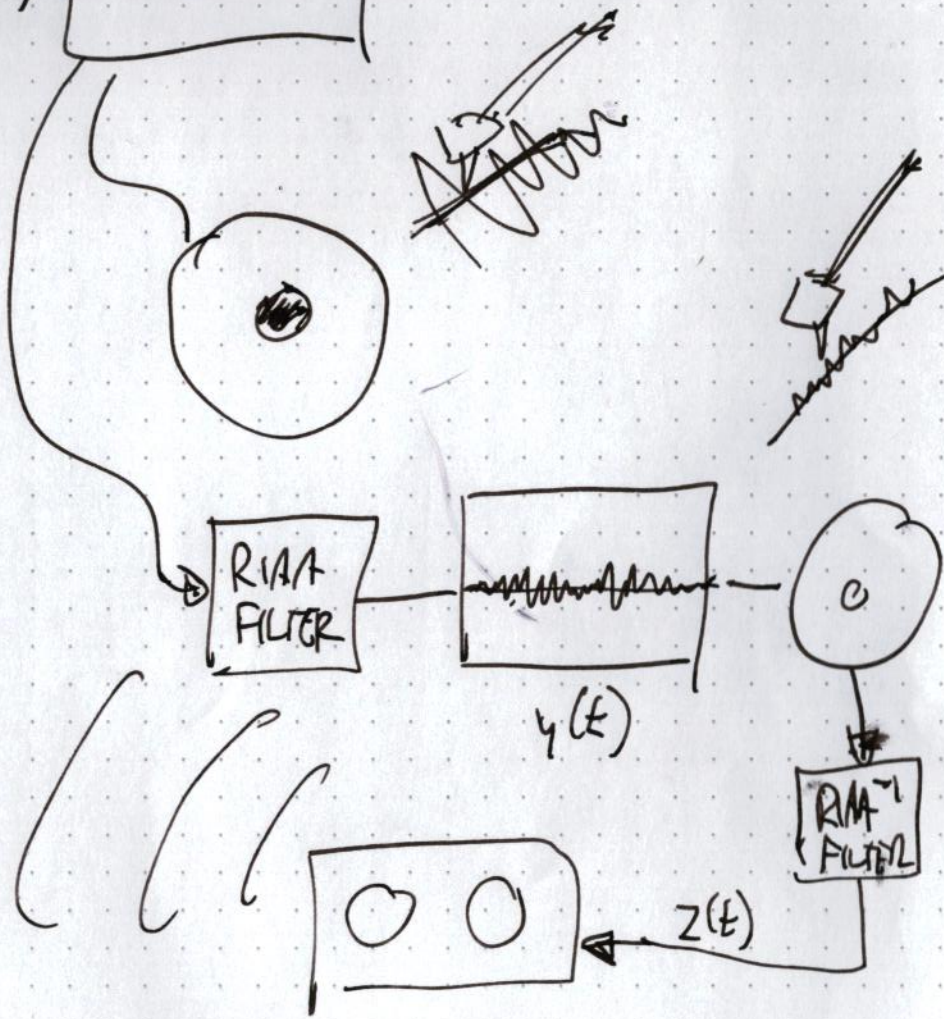
Non-Invertible

IF TWO DIFFERENT INPUTS CAUSE THE SAME OUTPUT, IT'S NON-INVERTIBLE



RIAA

INVERTIBLE SYSTEM



System Properties

Stable and Unstable Systems

Stability comes in two types: internal and external.

If every bounded input results in a bounded output, the system is externally stable. Also called **BIBO stable**.

Internal stability will be discussed later.

W03B

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BIBO STABILITY.

$$x(t) \rightarrow \boxed{S} \rightarrow y(t) = 0.5 (x(t) - x(-t))$$

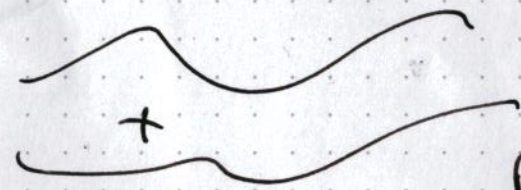
For an input we define:

$$|x(t)| \leq M_x < \infty$$

THIS DEFINES THE BOUNDS
OF THE INPUT.. -

Let's then PROVE $|y(t)|$ is ALWAYS BOUNDED...

$$|y(t)| = |0.5 (x(t) - x(-t))| \leq 0.5 (|x(t)| + |-x(-t)|)$$



What's the POSSIBLE BOUND THIS CAN BE?

So...

$$0.5 (|x(t)| + |-x(-t)|) \leq M_x < \infty$$

$$\rightarrow M_x$$

$$0.5 (M_x + M_x) = M_x.$$

∴ Bounded input produces Bounded output.

∴ BIBO STABLE

∴ THIS IS A NICE SYSTEM PROPERTY IF WE DON'T WANT THINGS TO GO TO INFINITY!
(current, speed, etc.)

System Response

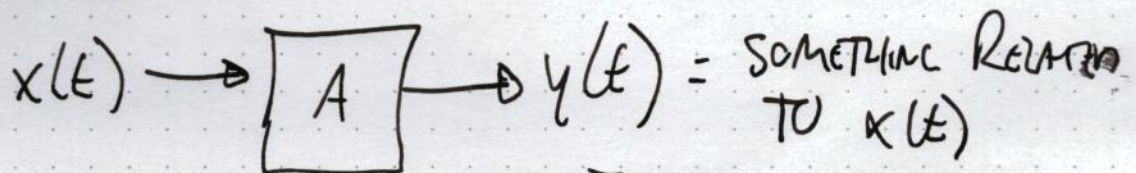
Linearity of Systems defined by Differential Equations

In the future we'll see that Laplace and Fourier transforms will help us easily characterize systems with outputs defined by differential equations.

A linear system response can be defined as:

total response = zero-input response + zero-state response

LINEARITY OF DE SYSTEMS



... JUST A RELATIONSHIP BETWEEN $x(t)$ & $y(t)$

$$\frac{dy(t)}{dt} + y^2(t) = x(t) \quad \rightarrow \text{CHECK FOR LINEARITY}$$

$$\left. \begin{array}{l} x_1(t) = K x(t) \\ y_1(t) = K y(t) \end{array} \right\} \begin{array}{l} \text{FOR LINEARITY} \\ \text{CHECK SCALING} \end{array}$$

For $x_1(t)$:

$$\frac{dy_1(t)}{dt} + y_1^2(t) = x_1(t) = K x(t)$$

From ABOVE

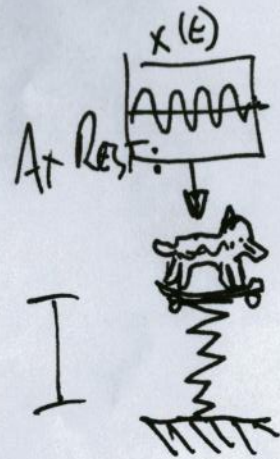
$$K x(t) = K \left(\frac{dy(t)}{dt} + y^2(t) \right) = \frac{dy_1(t)}{dt} + y_1^2(t)$$

Does this hold for $y_1(t) = K y(t)$?

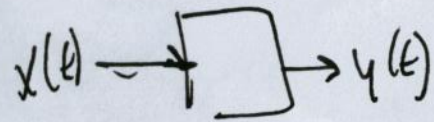
$$\frac{d(K y(t))}{dt} + (K y(t))^2 = K \frac{dy(t)}{dt} + K^2 y^2(t)$$

$\therefore$ THIS DOESN'T HOLD
SCALING ISN'T SATISFIED
SYSTEM IS NOT LINEAR.

TOTAL RESPONSE = ZERO-INPUT RESPONSE + ZERO-STATE RESPONSE

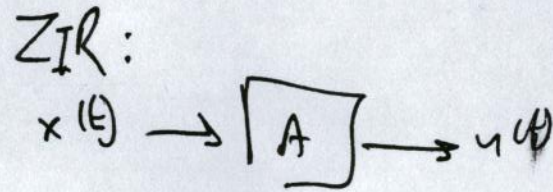


ZSR: ALL I.C.'S = 0
No stored energy, But can be driven by a source



$x(t)$ Position from eq DOAS

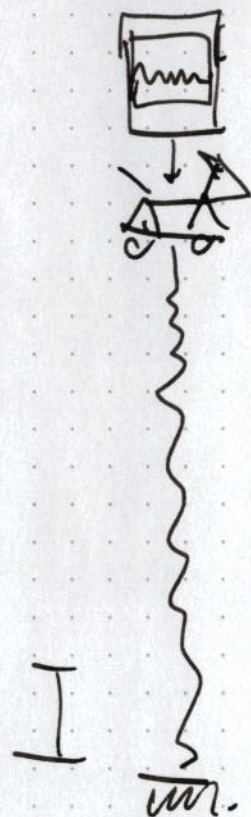
$y(t)$ Motion of DOAS



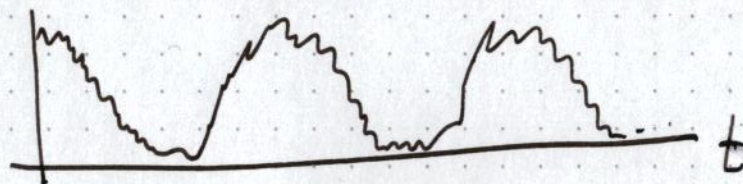
No input external $x(t)$,
But we can have an initial condition / stored energy.



DOAS



$ZSR + ZIR$



LINEARITY OF CONSTANT-COEFFICIENT LINEAR DE:

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

$$x(t) \rightarrow \boxed{S} \rightarrow y(t) = \text{SOLVE THIS DE}$$

LET'S ASSUME

$$x_1(t) \rightarrow y_1(t)$$
$$x_2(t) \rightarrow y_2(t)$$

$$\frac{dy_1(t)}{dt} + 3y_1(t) = x_1(t)$$

$$\frac{dy_2(t)}{dt} + 3y_2(t) = x_2(t)$$

CHECK FOR LINEARITY THE SAME WAY AS BEFORE
MULTIPLY BY $k_1 + k_2$

$$k_1 \frac{dy_1(t)}{dt} + 3k_1 y_1(t) = k_1 x_1(t)$$

$$k_2 \frac{dy_2(t)}{dt} + 3k_2 y_2(t) = k_2 x_2(t)$$

ADD THEM TOGETHER

$$\frac{d}{dt} [k_1 y_1(t) + k_2 y_2(t)] + 3 [k_1 y_1(t) + k_2 y_2(t)] = k_1 x_1(t) + k_2 x_2(t)$$

$$x(t) = k_1 x_1(t) + k_2 x_2(t)$$

$$y(t) = k_1 y_1(t) + k_2 y_2(t)$$

THIS SYSTEM IS LINEAR

THIS RESULT THEN GENERALIZES

$$a_0 \frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_N y(t) \\ = b_{N-m} \frac{d^m x(t)}{dt^m} + \dots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t)$$

IS A LINEAR SYSTEM

a_i & b_i ARE CONSTANTS OR FUNCTIONS OF TIME

WHY THIS MATTERS

IF OUR INPUT $x(t)$ CAN BE EXPRESSED
AS A SUM OF SIMPLER SIGNALS

$$x(t) = a_1 x_1(t) + a_2 x_2(t) + \dots + a_n x_n(t)$$

$$\rightarrow \boxed{S} \rightarrow y(t) = a_1 y_1(t) + a_2 y_2(t) + \dots + a_n y_n(t)$$

THE OVERALL OUTPUT $y(t)$ IS JUST THE
SUM OF THE RESPONSE TO THOSE SIMPLE INPUTS.

LINEAR SYSTEM RESPONSE

$$\overset{IN}{x(t)} \rightarrow [A] \rightarrow \overset{OUT}{y(t)}$$

ZERO INPUT RESPONSE

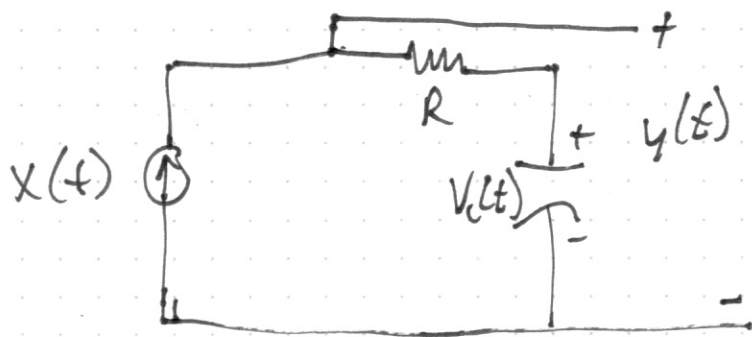
$$x(t) = 0 \text{ for } t \geq 0 \rightarrow [S] \rightarrow ZIR$$

ZERO STATE RESPONSE

$$\begin{array}{l} \text{INPUT } x(t) \text{ FOR } t \geq 0 \\ \text{WITH INITIAL CONDITIONS} = 0 \end{array} \rightarrow [S] \rightarrow ZSR$$

$$\text{TOTAL RESPONSE} = ZIR + ZSR$$

ALLOWS US TO DECOMPOSE AN OUTPUT RESPONSE INTO THE PARTS DUE TO THE SIGNAL + THE INITIAL CONDITIONS



$$y(t) = \underbrace{V_c(0)}_{ZIR} + \underbrace{Rx(t) + \frac{1}{C} \int_0^t x(\tau) d\tau}_{ZSR}$$

$$ZIR \quad \text{IF } x(t) = 0 \text{ for } t \geq 0 \quad y(t) = V_c(0)$$

$$ZSR \quad \text{IF } V_c(0) = 0 \text{ AT } t_0 \quad y(t) = Rx(t) + \frac{1}{C} \int_0^t x(\tau) d\tau$$